The conjectured generating function for OEIS A277866, proved

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Abstract

OEIS A277866 counts arrays of a fixed width under a local condition, and carries a conjectured generating function — and nothing else. It is true. The count is a walk count in a finite digraph on $S = 8$ vertices, so its generating function is a rational function whose numerator and denominator have degrees bounded by S; the conjectured expression is another rational function of known degree; and two rational functions of bounded degree agree as soon as enough coefficients of their expansions do. Comparing 17 coefficients in exact arithmetic therefore settles the conjecture, rather than testing it.

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1 The conjecture

OEIS A277866 is "Decimal representation of the x-axis, from the left edge to the origin, of the n-th stage of growth of the two-dimensional cellular automaton defined by "Rule 3", based on the 5-celled von Neumann neighborhood.". It has offset 0 and begins

$$1, 1, 1, 13, 1, 61, \ \ldots$$

The entry states, as a conjecture contributed by a reader and never marked settled:

G.f.: (1+x-4*x^2+8*x^3) / ((1-x)*(1+x)*(1-2*x)*(1+2*x)).

As of the "Last modified" line on the live entry (Nov 05 2025 15:22:35, revision 13) this is still recorded as empirical, and nothing on the entry records it as proved. The entry carries no conjectured recurrence, which is why this generating function is the whole of what is open here.

2 The value is a certified growth

The entry reads an axis or a diagonal of the automaton at stage n as a string of digits and takes its numeric value. Nothing is being counted, so there is no digraph. What settles the sequence is that the CONFIGURATION is periodic in time:

$$C(n+2) \ = \ C(n) \qquad \text{as configurations of the whole plane, for every } n \geq 0.$$

This was checked directly, and one instance of it is all that is needed: the automaton is a deterministic map on configurations, so $C(n_0+2) = C(n_0)$ forces $C(n_0+k\cdot 2) = C(n_0)$ for every k, and likewise in each residue class. No locality argument and no induction over boundary windows is involved.

Getting the background right is the whole of the check. A rule taking an empty von Neumann neighbourhood to a live cell flips the ENTIRE plane, and rule 3 is read in that light: comparing two stages by padding the smaller with zeros would compare two different things.

The reading follows at once. The axis at stage n is the window $|x| \le n$ of a configuration that is not changing, so the window widening by one cell on each side per step gives

$$w(n+2) \;=\; L_r + w(n) + R_r$$

with L_r and R_r fixed words depending only on the residue r of n modulo 2 — an identity that is now a consequence rather than an observation. Concatenation is multiplication by a power of the base B and addition, so along a class

$$a(n+2) \;=\; v(L_r)\, B^{|w(n)|+|R_r|} + a(n)\, B^{|R_r|} + v(R_r).$$

3 The bound

In the shift $T = S^2$ that is a first-order recurrence with a geometric forcing term and a constant, killed by $(T - B^{|R_r|})(T - B^{|L_r|+|R_r|})(T-1)$. Taking the DISTINCT roots that occur across the residue classes, pulling each back to $S^2 - \lambda$, and allowing the pre-period to raise the numerator's degree gives a monic annihilator of order $S = 8$.

The family this entry belongs to is read by one model, described by its author as: "The x-axis and diagonal of a two-dimensional cellular automaton, read as a number."

The model reproduces all 43 terms the entry publishes, exactly, in integer arithmetic: it is built by reading the entry's own English, and a misreading gives different counts, so this ties the model to the entry and not merely to the arithmetic.

4 Its generating function is rational, with bounded degree

Lemma 1. *$A(x) = \sum_{n\ge0} a(n)x^n = N_1(x)/D_1(x)$ where $D_1(x)$ has degree at most $S = 8$ and $D_1(0) = 1$, and $\deg N_1 < S + 0$.*

Proof. The section above derives a MONIC linear recurrence of order S that a provably satisfies from some index onwards; write its characteristic polynomial as $z^S - \sum_{j<S} \alpha_j z^j$ and put $D_1(x) = 1 - \sum_{j<S} \alpha_j x^{S-j}$, the same polynomial written backwards, so $\deg D_1 \le S$ and $D_1(0) = 1$. Multiplying the power series $A(x)$ by $D_1(x)$ annihilates every coefficient at which the recurrence is in force, so $N_1 = AD_1$ is a polynomial, and the indices it can be supported on are the offset together with the finitely many below the point where the recurrence takes hold, all of them less than $S + 0$. No matrix is involved: the model here is not a walk and has no adjacency matrix, and the bound comes from the recurrence the model itself supplies. □

Theorem 2. *The conjectured generating function is $A(x)$.*

Proof. Write the conjecture as N_2/D_2 with $\deg N_2 = 3$ and $\deg D_2 = 4$; its expansion as a power series exists because $D_2(0) \ne 0$. Put

$$R(x) \;=\; N_1(x)D_2(x) - N_2(x)D_1(x),$$

a polynomial of degree at most 11 by Lemma 1. The difference $A - N_2/D_2$ equals $R/(D_1D_2)$, and $D_1(0)D_2(0) \ne 0$, so if the two power series agree in every coefficient up to x^{11} then R has a zero of order greater than its own degree at the origin, which forces $R \equiv 0$ and hence $A = N_2/D_2$ identically.

The coefficients of A were produced from the model by repeated matrix–vector products in exact integer arithmetic, those of N_2/D_2 by exact rational long division, and 17 of them — more than the $11 + 1$ the argument requires — agree. □

5 A recurrence the entry does not state

The denominator of a rational generating function is a linear recurrence written backwards, so the theorem above yields one for free. With $D_2(x) = 1 - \sum_{i\geq 1} c_i x^i$ after normalising $D_2(0) = 1$, comparing coefficients of x^n in $A(x)D_2(x) = N_2(x)$ gives

$$a(n) \;=\; \sum_{i\geq 1} c_i\, a(n-i) \qquad \text{for every } n > 3,$$

a constant-coefficient linear recurrence of order 4, valid past the degree of the numerator. Its coefficients are the integers $c_1, \ldots, c_4$ read off D_2, the first few being $0,\ 5,\ 0,\ -4$. The entry records no recurrence, so this is not a restatement of anything already there.

6 Verification

Three checks, each independent of the algebra above.

First, the model reproduces every one of the 43 published terms, with the index shift read off the entry's offset rather than fitted.

Second, the arrays themselves were enumerated for the smallest shapes the entry counts, directly from its English and with no automaton, and the counts agree. Writing that check is where the risk actually sits: an earlier version of it dropped a clause from the entry's name in three cases and disagreed with the model, and in all three the check was wrong and the model was right.

Third, the coefficient comparison was carried out over the whole range required by the degree bound rather than on a prefix, in exact arithmetic throughout.

References

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